EVENTEASE

Complete Project Documentation

Features, Roles and Workflows

Prepared from the current EventEase application source code

1. Project Overview

EventEase is a cross-platform event-management application built with Flutter. It brings event discovery, registration, ticketing, event operations, administrative control, and post-event engagement into one system.

The platform supports three distinct user roles:

- **Attendee:** discovers events, registers or purchases a ticket, attends the event, and completes post-event actions.
- **Organizer:** creates and operates events, manages participants, verifies attendance, and manages event content.
- **Super Admin:** controls platform users, reviews approvals, manages events, moderates content, and produces reports.

The application uses Firebase Authentication for login, Cloud Firestore for application data, Firebase Storage for media, and Provider for state management. It is configured for Android, iOS, web, Windows, macOS, and Linux.

2. Roles and Their Scenarios

2.1 Attendee Scenario

An attendee creates an account, logs in, browses approved events, applies search and filters, and opens an event's detail page. They can save events as favorites, register for a free event, or use **Buy Now** for a paid event. After successful registration, the attendee receives a QR event pass.

On the event day, the attendee presents this QR pass to the organizer. The organizer scans it and marks verified attendance. After the event is completed, the attendee can open the completed event in **My Tickets**, submit a review, and-when attendance has been verified-generate, download, or share a participation certificate.

2.2 Organizer Scenario

An organizer registers for an organizer account. The account initially stays pending until it is approved by the Super Admin. Once approved, the organizer can create an event by entering its content, schedule, capacity, price, rules, contact information, image, and map location.

Every newly created event is submitted for administrative approval. Once approved, it becomes available to attendees. The organizer can then monitor registrations, view event participants, scan attendee QR passes to record attendance, send event announcements, view attendee feedback, edit or cancel events, and upload event-gallery images.

2.3 Super Admin Scenario

The Super Admin manages the entire EventEase platform. From the dashboard, the admin can review pending organizer accounts and pending event submissions. The admin can approve or reject either item before it goes live.

The admin can also search users, suspend or reactivate accounts, view and edit event information, change event status, delete events when required, moderate flagged gallery content, and create an administrative PDF report with platform metrics and event information.

3. Common Features

3.1 Onboarding and Authentication

- Onboarding introduces the purpose of EventEase and the role-based platform experience.
- Users can register as either an Attendee or Organizer.

- Login supports email/password authentication and Google sign-in.
- The system reads the signed-in user's role and opens the correct dashboard automatically.
- Organizer accounts require Super Admin approval before the organizer can use the dashboard.
- A user can log out and sign in again; the application retains role context for a smoother session restore.

3.2 Profile and Settings

All roles have a profile and settings area. Users can view and update their personal information and profile image. The profile image can be selected from the device gallery or camera. General settings, application preferences, and logout are available from the same role-specific dashboard environment.

3.3 About Us and Contact Us

- **About Us** explains EventEase, its purpose, its three roles, and the platform's main value proposition.
- **Contact Us** lets a user send a question, feedback item, or event inquiry to the platform team. Submitted messages are stored for follow-up.

4. Attendee Features

4.1 Discover Events

The attendee's home area shows approved events in a card-based discovery experience. Event cards provide quick access to the event image, category, date, venue, price, available seats, and organizer information. The attendee can open any card to view full details.

4.2 Search and Advanced Filters

Attendees can search approved events and narrow results using the following filters:

- Event title, description, organizer, or general keyword.
- Category.
- Specific date.
- Location.
- Only events with seats still available.
- Maximum price.

The user can clear all filters at once to return to the complete approved-event list.

4.3 Event Details

The event detail page presents all important event information in one place:

- Event title, poster/image, category, date, start time, end time, and venue.
- Price and remaining-seat information.
- Organizer details.
- Full event description.
- Rules and guidelines, when provided by the organizer.
- Contact information, when provided by the organizer.

- Venue map preview and access to the full map screen.
- Favorite/unfavorite action.
- Registration action for free events or **Buy Now** action for paid events.

4.4 Favorites

An attendee can mark an event as a favorite from the event detail page. Favorite events appear in a dedicated Favorites section, allowing the attendee to revisit them easily. The attendee can remove a favorite at any time.

4.5 Registration and QR Event Pass

For a free event, the attendee selects **Register Now**. For a paid event, the attendee selects **Buy Now** and completes the purchase flow described in the Extra Features section below.

The system prevents duplicate active registrations and prevents registration when the event is sold out. A successful registration creates an attendee event record and a unique QR code. The attendee can open **My Tickets** and tap an upcoming event to display this QR pass at check-in.

The attendee may cancel an active registration. When cancellation succeeds, the ticket is marked cancelled and the event seat is returned to the available-seat count.

4.6 My Tickets / My Events

The attendee's events are separated into three easy-to-understand tabs:

- **Upcoming:** currently active registered events. Each card opens the attendee QR pass.
- **Completed:** attended or past events. This area offers review submission and, after verified attendance, certificate access.
- **Cancelled:** events or registrations that are no longer active.

4.7 Event Gallery

Attendees can browse images uploaded to the event gallery. The gallery supports a full-screen image experience and allows the attendee to view event memories after an event.

4.8 Feedback and Rating

For a completed event, an attendee can submit feedback. The feedback form includes:

- A 1-to-5 star rating.
- Optional feedback tags.
- Written comments.

The system prevents the same attendee from submitting duplicate feedback for the same event. The organizer can later view this feedback for that event.

5. Organizer Features

5.1 Organizer Dashboard

The organizer dashboard gives a focused overview of the organizer's own events, participant activity, and feedback. It helps the organizer identify event status, registrations, and operational actions without accessing any other organizer's data.

5.2 Create an Event

An organizer can create an event with the following information:

- Title, category, description, and event image.
- Event date, start time, and end time.
- Venue/location and selected geographic coordinates.
- Maximum participants and available seats.
- Free or paid event price.
- Event rules and guidelines.
- Contact information.

The organizer can choose an image from the device gallery or camera. The location picker enables the organizer to choose the venue directly on a map and stores both the readable address and coordinates.

New events are created in a pending state and require Super Admin approval before attendees can discover or register for them.

5.3 Edit, Cancel, and Manage Own Events

The organizer can edit event information when changes are needed. Important event changes are sent through the approval workflow. The organizer can also cancel an event. Cancelled events are protected from further attendee check-ins and from new announcements.

5.4 Participants Management

For every organizer-owned event, the organizer can open the participant list. This provides a direct view of registered attendees and their registration status, enabling the organizer to manage attendance and event readiness.

5.5 QR Attendance Scanner

The organizer can use the device camera to scan an attendee's QR event pass. The scanner:

- Confirms that the scanner is being used by an authenticated organizer.
- Confirms that the QR pass belongs to a valid registration.
- Ensures the organizer owns the selected event.
- Blocks check-in for a cancelled event.
- Stops duplicate attendance marking.
- Records the attendee as **Present** after a successful scan.

Verified attendance is the basis for an attendee's certificate eligibility.

5.6 Announcements

An organizer can select one of their events and prepare an announcement with a title and message. The announcement is saved against the relevant event and made available to that event's registered attendees through the application's update area. Announcements cannot be issued for a cancelled event.

5.7 Feedback Review

The organizer can view event-specific feedback submitted by attendees. The feedback area shows ratings, attendee comments, and summary information so that the organizer can evaluate the event experience and improve future events.

5.8 Gallery Upload

For an approved event, the organizer can upload gallery photographs from the device gallery or camera. Uploaded images are associated with the chosen event and then become viewable in the attendee gallery. The system does not allow gallery uploads for an event that has not been approved.

6. Super Admin Features

6.1 Admin Dashboard

The Super Admin dashboard provides an overall operational view of EventEase. It includes counts and quick actions for users, active events, pending approvals, flagged content, registrations, and feedback performance.

6.2 Organizer Account Approval

Every newly registered organizer is shown in the approval queue. The Super Admin can review the organizer request and either:

- **Approve:** activates the organizer account so that the organizer can access organizer tools.
- **Reject:** declines the request so the account is not activated for organizer use.

6.3 Event Approval

New organizer-created events also enter the approval queue. The Super Admin reviews them before they become visible to attendees. The admin can approve or reject each event submission.

6.4 User Management

The Super Admin can view platform users, search them by name, email, or role, and review role and account-status information. The admin can suspend an account when needed and reactivate it later.

6.5 Event Management

The Super Admin can view all events, inspect full event details, edit event data, change event status, and delete an event when required. Event information managed here includes schedule, category, location, price, capacity, available seats, rules, contact information, and status.

6.6 Content Moderation

Flagged gallery content appears in the moderation queue. The Super Admin can review a flagged item and either ignore the flag or remove the gallery item from the platform.

6.7 Reports

The Super Admin can create an EventEase PDF report. The report includes a system overview, user count, active-event count, estimated revenue, pending approvals, flagged-content count, and a list of events with key metrics. It can be exported through the device's PDF/print flow.

7. Extra Features

This section documents the requested additional features separately.

7.1 Map and Directions - Organizer and Attendee

Organizer side

While creating or editing an event, the organizer can open the location picker, choose the event venue on the map, and save the selected location. The system stores the venue address along with latitude and longitude. This makes the event location precise rather than relying only on manually typed text.

Attendee side

The attendee can open the map from the event detail screen. The map shows:

- The event venue marker.
- The attendee's current location, after location permission is granted.
- A route from the attendee's current position to the event venue.
- The remaining distance to the destination.

When the attendee presses **Start Route**, the application enters route mode, centers the map around the user, tracks movement, and updates the visible route as the attendee travels. The attendee can use the **My Location** control to center the map on their current position and can stop the route at any time. On arrival near the event location, the app confirms that the destination has been reached.

The map also provides a dedicated **Google Maps** button on the right side. When pressed, it opens the official Google Maps application when available; otherwise, it opens Google Maps externally in a browser. The destination coordinates are passed to Google Maps so the attendee can continue with full turn-by-turn navigation there.

7.2 Buy Now - Paid Event Purchase

The **Buy Now** button is available only to the Attendee on a paid event's detail screen. A free event shows **Register Now** instead.

Buy Now flow:

1. The attendee opens a paid event's detail screen and presses **Buy Now**.
2. The attendee chooses a payment method: Easypaisa, JazzCash, PayPal, or Cash.
3. For Easypaisa and JazzCash, the attendee enters a valid Pakistani mobile account number. For PayPal, the attendee enters a valid email address. Cash does not require account details.
4. The attendee reviews and confirms the payment details.
5. The system creates a confirmed registration, records the payment method, marks the payment status as paid, records the amount, generates the QR pass, and reduces the available-seat count by one.

The purchase is blocked when the event is sold out or the attendee already has an active registration for the event.

7.3 Certificate Generation - Attendee

The certificate feature is available to an Attendee after the event is completed **and** the organizer has verified that attendee's attendance by scanning the QR pass.

Certificate flow:

1. The attendee opens **My Tickets** and selects the **Completed** tab.
2. If verified attendance exists for that event, the **Certificate** option appears on the event card.
3. The attendee opens the certificate preview, which includes the attendee name, event title, event date, organizer name, and a unique certificate ID.
4. The attendee can choose **Download PDF** to generate and save the certificate as a PDF.
5. The attendee can also choose **Share Certificate** to generate the PDF and share it using the device's available sharing options.

The certificate is therefore not available merely because a ticket was purchased; it is unlocked only after verified presence at the completed event.

8. Main End-to-End Workflows

8.1 Event Publication Workflow

1. Organizer account is registered.
2. Super Admin approves the organizer account.
3. Organizer creates an event and selects the event location on the map.
4. The event enters the pending approval queue.
5. Super Admin reviews and approves the event.
6. The event appears in the attendee discovery area.

8.2 Attendee Registration and Attendance Workflow

1. Attendee finds an approved event through discovery or search.
2. Attendee opens the event detail page and reviews details, map, rules, and price.
3. Attendee uses Register Now for a free event or Buy Now for a paid event.
4. The system creates a confirmed ticket with a unique QR pass.
5. On event day, the attendee presents the QR pass.
6. Organizer scans the pass and marks attendance as Present.
7. After event completion, the attendee can submit feedback and generate a certificate if attendance was verified.

8.3 Event Gallery Workflow

1. Organizer selects an approved event.
2. Organizer chooses a photo from the gallery or camera.
3. Organizer uploads the photo to that event's gallery.
4. Attendees open Event Gallery to view the uploaded photos, including full-screen viewing.

9. Core Data Managed by the System

9.1 User Data

User profiles store identifiers, name, email, phone, role, profile image, and account status. Roles determine which dashboard and permitted features are available.

9.2 Event Data

Each event stores title, description, category, price, organizer and organizer ID, image, location, date, start and end time, rules, contact information, maximum participant count, available-seat count, approval status, and optional latitude/longitude coordinates.

9.3 Registration Data

Each ticket/registration stores the event ID, attendee ID, attendee name, email, registration date, registration status, QR code value, payment method, payment status, and amount paid.

9.4 Attendance, Feedback, and Gallery Data

Attendance records establish verified event presence. Feedback records capture the attendee's rating, tags, comment, and event association. Gallery records associate uploaded photos with their events and can be flagged for Super Admin moderation.

10. Important Business Rules

- Only approved events are displayed for attendee discovery and registration.
- Organizer accounts require Super Admin approval before organizer operations are available.
- New organizer events require Super Admin approval before publication.
- An attendee cannot keep more than one active registration for the same event.
- A registration cannot be made when all event seats are occupied.
- Cancelling a registration returns a seat to the event's available-seat count.
- Only the event's owning organizer can record attendance for that event.
- A QR pass cannot be used to mark attendance more than once.
- Attendance cannot be recorded for a cancelled event.
- Only approved events accept gallery uploads.
- An attendee can submit feedback once per event.
- A certificate appears only for a completed event with verified attendance.

11. Technology Summary

- **Frontend:** Flutter and Dart.
- **Authentication:** Firebase Authentication, including Google sign-in.
- **Database:** Cloud Firestore.
- **Media:** Firebase Storage and image upload support.
- **State management:** Provider.

- **QR tickets and scanning: qr_flutter** and **mobile_scanner**.
- **Maps and location:** Flutter Map, OpenStreetMap tiles, OSRM routing, Geolocator, and external Google Maps launch support.
- **PDF output:** PDF and Printing packages for administrative reports and attendee certificates.
- **Supported targets:** Android, iOS, web, Windows, macOS, and Linux.

This documentation is based on the current EventEase source code.